MATH 1400 Quiz 2 Name: ____________________________

Date: ________________ Section: ___________

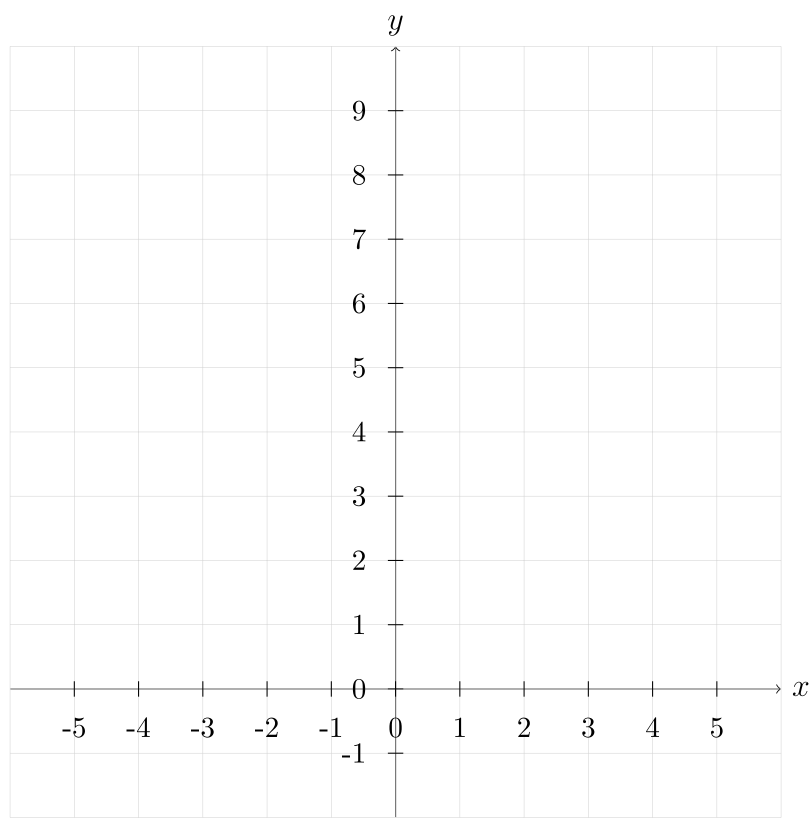
1. **Piecewise Function.** Consider

$$f(x) = \begin{cases} -2x+1, & -3 \le x < 1, \\ 2, & x = 1, \\ x^2, & x > 1. \end{cases}$$

(a) Compute $f(-2)$, $f(1)$, and $f(2)$.

(b) Find the domain of f. Then find all intercepts (x- and y-intercept).

(c) Sketch the graph of f on the grid. Clearly mark open/closed circles.



2. **Reciprocal Function.** Let $g(x) = \dfrac{1}{x}$.

   (a) Is g even, odd, or neither? Justify using $g(-x)$.

   (b) State the interval(s) where g is increasing/decreasing.

3. **Square Function.** Let $h(x) = x^2$.

   (a) Is h even, odd, or neither? Justify briefly.

   (b) State where h is increasing and where it is decreasing.

   (c) Does h have an absolute maximum or absolute minimum? If so, state the value and where it occurs.